Beyond Asymptotics: Finite-Blocklength Linear Coding with Strictly Causal Side Information

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***Abstract*—THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD. Channels with strictly causal interference side information arise when the encoder only has access to interference side information up to time step $i-1$ at current step i. Despite their prevalence in sequential physical systems, coding designs for such channels remain largely unexplored. Traditional dirty-paper precoding methods are infeasible here, as they cannot eliminate interference solely based on past side information. To address the gap, this paper proposes a linear coding scheme for Gaussian channels with strictly causal interference side information by exploiting its connection to feedback channel coding. The proposed scheme achieves the channel capacity when the interference channel is noiseless. In the presence of both interference and noise, although the achievable rate asymptotically vanishes over infinite blocklengths, finite blocklength analysis indicates that the proposed linear scheme remains highly beneficial in binary modulation and high-interference regimes.**

***Index Terms*—Strictly-causally known side information, linear coding, finite blocklength analysis, asymptotic rate.**

I. Introduction

Additive Gaussian channels with interference state information at the encoder are categorized based on the encoder's knowledge at time step i over a total of N channel uses: non-causal (knowledge of the full sequence $\{1, \cdots, N\}$), causal (knowledge up to step i), and strictly-causal (knowledge up to step $i-1$). Remarkable progress has been made in channel coding for non-casually known interference. Costa [1] showed that when interference is known non-causally at the encoder, the channel capacity is identical to that of an interference-free channel. This landmark finding led to the development of dirty paper coding (DPC) [1], where the encoder pre-compensates for interference such that transmitted codewords are effectively orthogonal to interference. Erez *et al.* [2] subsequently utilized lattice quantization to extend DPC to the causal setting, showing that capacity loss vanishes as the lattice dimension approaches infinity. In contrast, the strictly-causal setting is far more constrained. Shannon [3] proved that access to independently and identically distributed (i.i.d.) strictly-causally known side information does not increase channel capacity. Traditional DPC-based schemes are inapplicable in this setting, as they require interference side information available causally. As a result, practical coding for strictly-causally known interference channels remains an open problem requiring fundamentally distinct design principles.

In this paper, we propose a linear coding scheme for additive Gaussian channels with strictly causal interference side information at the encoder. The underlying channel model motivates a feedback-oriented interpretation: since both settings rely exclusively on past side information, the strictly-causally known interference at the encoder exhibits fundamental similarities to feedback. Specifically, since the encoder observes the past interference sequence but lacks access to the additive channel noise, the system can be viewed as a form of a noisy feedback channel [4]–[7], where the encoder possesses partial side information. Our design principle centers on iteratively refining the interference estimate by leveraging the histroty of available side information.

The linear code design is first formulated as a signal-to-interference-plus-noise ratio (SINR) maximization problem subject to a power constraint, where the encoder and decoder parameters are derived via its equivalent power-minimization dual. Although the proposed linear scheme is capacity-achieving for noiseless interference channels, its achievable rate vanishes asymptotically in the presence of additive noise. Because asymptotic limits do not capture finite-latency behavior, we utilize the finite-blocklength framework [8] to identify the operational regimes where the scheme remains effective. Our analysis shows that for a message size $M = 2$ and arbitrary blocklength N, the proposed scheme achieves a strictly positive rate for a target error probability ϵ. In contrast, such performance is impossible for an encoder without side information. For sufficiently large blocklength N and $M \geq 2$, our analysis reveals that the proposed scheme is advantageous in high-interference regimes. Specifically, when the interference power exceeds the noise power, we demonstrate the existence of a feasible target error probability ϵ for which the scheme achieves a rate $R > R_{\text{no-SI},\epsilon}$, where $R_{\text{no-SI},\epsilon}$ represents the maximum rate achievable with error probability ϵ without side information. The full version of this paper with the appendix is available at [9].

Notation. $\mathcal{N}(\mu, \sigma^2)$ denotes a Gaussian distribution with mean μ and variance σ^2. We write $f(x) = o(g(x))$ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$, and $f(x) = O(g(x))$ if $\limsup_{x\to\infty} \frac{|f(x)|}{g(x)} < \infty$.

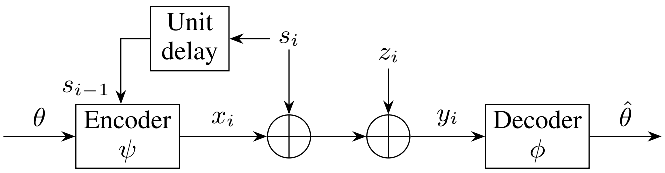

Figure 1: System model of the Gaussian channel with strictly causal interference side information at the encoder.

II. System model and problem statement

The system model of the additive Gaussian channel with strictly causal interference side information at the encoder is shown in Figure 1. The message symbol θ is uniformly drawn over a rate R (bits/channel use) message set $\mathcal{C} = \{\theta_1, \cdots, \theta_M\}$ with $E[\theta^2] = 1$ and the message set size $M = 2^{NR}$, where the blocklength N is utilized for the transmission of one symbol $\theta \in \mathcal{C}$. The encoder has access to the strictly-causal interference side information $s_1^{i-1} = (s_1, \cdots, s_{i-1})$ at the i-th channel use, where the interference $s_i \sim \mathcal{N}(0, \sigma_s^2)$ is i.i.d. The i-th channel input x_i is of the form, $x_i = \psi(\theta, s_1^{i-1})$, where ψ is the encoding function depending on the past side information s_1^{i-1} and message symbol θ. The i-th channel output is then given by

$$y_i = x_i + s_i + z_i, \quad i = 1, \cdots, N, \tag{1}$$

where the additive i.i.d. noise $z_i \sim \mathcal{N}(0, \sigma^2)$ is assumed to be independent of s_i. The decoding process follows $\hat{\theta} = \phi(y_1, \cdots, y_N)$, where ϕ is the decoding function to estimate transmitted symbol $\hat{\theta}$ from y_1^N. We put power constraint on x_i such that $\sum_{i=1}^N \mathbb{E}[x_i^2] \le NP$.

The encoder ψ is linear parameterized by a vector $\mathbf{g} \in \mathbb{R}^{N\times 1}$ to encode θ and a strictly lower-triangular matrix $\mathbf{F} \in \mathbb{R}^{N\times N}$ with zero diagonal to encode strictly causal interference. The linearly encoded transmit signal vector is

$$\mathbf{x} = \mathbf{g}\theta + \mathbf{F}\mathbf{s}, \tag{2}$$

where $\mathbf{s} = [s_1, \ldots, s_N]^T \in \mathbb{R}^{N\times 1}$. The channel output is then

$$\mathbf{y} = \mathbf{x} + \mathbf{s} + \mathbf{z} = \mathbf{g}\theta + (\mathbf{I} + \mathbf{F})\mathbf{s} + \mathbf{z}. \tag{3}$$

The decoder ϕ employs a linear combiner $\mathbf{w} \in \mathbb{R}^{N\times 1}$ to estimate θ from $\mathbf{y}$,

$$\hat{\theta} = \mathbf{w}^T\mathbf{y} = \mathbf{w}^T\mathbf{g}\theta + \mathbf{w}^T\big((\mathbf{F} + \mathbf{I})\mathbf{s} + \mathbf{z}\big). \tag{4}$$

Given the linear models in (2)-(4), the goal is to optimize $\mathbf{g}$, $\mathbf{F}$, and $\mathbf{w}$ to maximize the post-processing SINR, $\text{SINR} = \frac{\mathbb{E}[|\mathbf{w}^T\mathbf{g}\,\theta|^2]}{\mathbb{E}[|\mathbf{w}^T((\mathbf{F}+\mathbf{I})\mathbf{s}+\mathbf{z})|^2]}$ under a transmit power constraint. The SINR maximization problem is then given by

$$\begin{aligned} \mathcal{K}: \quad & \max_{\mathbf{g},\mathbf{F},\mathbf{w}} \ \text{SINR} = \frac{|\mathbf{w}^T\mathbf{g}|^2}{\mathbf{w}^T\mathbf{C}\mathbf{w}} \\ \text{subject to} \quad & \sum_{i=1}^N \mathbb{E}[x_i^2] = \|\mathbf{g}\|^2 + \sigma_s^2\,\text{tr}(\mathbf{F}\mathbf{F}^T) \le NP, \end{aligned} \tag{5}$$

where $\mathbf{C} = \sigma_s^2(\mathbf{F}+\mathbf{I})(\mathbf{F}+\mathbf{I})^T + \sigma^2\mathbf{I}$ is the interference-plus-noise covariance matrix and $\text{tr}(\cdot)$ denotes the matrix trace.

This linear encoding and decoding framework is fundamentally rooted in the principles of feedback communication [4], [5] where the encoder utilizes past channel states to refine current transmission.

III. Linear Coding with Strictly Causal Interference Side Information

The proposed linear approach is inspired by the linear coding designs for noisy feedback channels established in [5]–[7], which we adapt here to the unique context of strictly-causal interference by treating past side information s_1^{i-1} as a form of noisy feedback.

A. Linear Decoder Design

We define $\mathbf{g} = \mathbf{C}^{1/2}\mathbf{q}$, where $\mathbf{q} \in \mathbb{R}^{N\times 1}$. Then, by the Cauchy–Schwarz inequality, the following holds

$$\text{SINR} = \frac{\left|\mathbf{w}^T\mathbf{C}^{1/2}\mathbf{q}\right|^2}{\mathbf{w}^T\mathbf{C}\mathbf{w}} \le \frac{\left\|\mathbf{w}^T\mathbf{C}^{1/2}\right\|^2\|\mathbf{q}\|^2}{\mathbf{w}^T\mathbf{C}\mathbf{w}} = \|\mathbf{q}\|^2. \tag{6}$$

The upper bound is achieved with equality by choosing $\mathbf{q} = \alpha\mathbf{C}^{1/2}\mathbf{w}$ for $\alpha \ne 0$, i.e., $\mathbf{w} = \frac{1}{\alpha}\mathbf{C}^{-1/2}\mathbf{q} = \frac{1}{\alpha}\mathbf{C}^{-1}\mathbf{g}$. The unbiased estimate of $\hat{\theta}$ is obtained with $\alpha = \mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}$, where

$$\hat{\theta}_N = \mathbf{w}^T\mathbf{y} = \frac{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{y}}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}, \quad \mathbf{w} = \frac{\mathbf{C}^{-1}\mathbf{g}}{\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g}}. \tag{7}$$

Thus, the post-processing SINR is $\mathbf{g}^T\mathbf{C}^{-1}\mathbf{g} = \|\mathbf{q}\|^2$, which is maximized when $\mathbf{g}$ is the eigenvector associated with minimum eigenvalue of $\mathbf{C}$, i.e., $\lambda_{\min}(\mathbf{C})$. From the equality $\mathbf{g} = \mathbf{C}^{1/2}\mathbf{q}$, it is immediately clear that

$$\mathbf{g} = \sqrt{\lambda_{\min}(\mathbf{C})}\,\mathbf{q}. \tag{8}$$

Now, the transmit power constraint in (5) can be rewritten as $\mathbb{E}[\mathbf{x}^T\mathbf{x}] = \mathbf{q}^T\mathbf{C}\mathbf{q} + \sigma_s^2\text{tr}(\mathbf{F}\mathbf{F}^T) = \sigma_s^2\|(\mathbf{I}+\mathbf{F})^T\mathbf{q}\|^2 + \sigma^2\|\mathbf{q}\|^2 + \sigma_s^2\text{tr}(\mathbf{F}\mathbf{F}^T)$. Then, the linear coding parameters $\mathbf{q}$ and $\mathbf{F}$ are designed by considering an alternative SINR-constrained power minimization problem,

$$\begin{aligned} \mathcal{J}: \min_{\mathbf{q},\mathbf{F}} \ & \mathbb{E}[\mathbf{x}^T\mathbf{x}] = \sigma_s^2\|(\mathbf{I}+\mathbf{F})^T\mathbf{q}\|^2 + \sigma^2\|\mathbf{q}\|^2 + \sigma_s^2\text{tr}(\mathbf{F}\mathbf{F}^T) \\ \text{subject to} \quad & \|\mathbf{q}\|^2 = \eta, \end{aligned} \tag{9}$$

where $\|\mathbf{q}\|^2 = \eta$ is a fixed SINR constraint. It is not difficult to show that the max-SINR problem $\mathcal{K}$ and the min-power problem $\mathcal{J}$ are equivalent. In what follows, we focus on optimizing $\mathbf{q}$ and $\mathbf{F}$ by solving $\mathcal{J}$.

B. Linear Encoder Design

We denote $[0, \ldots, 0, \mathbf{f}_i^T]^T$ as the ith column of the strictly lower triangular matrix $\mathbf{F}$, where $\mathbf{f}_i = [f_{i+1,i}, \ldots, f_{N,i}]^T \in \mathbb{R}^{(N-i)\times 1}$ and let $\mathbf{u}_i = [q_{i+1}, \ldots, q_N]^T \in \mathbb{R}^{(N-i)\times 1}$ with $\mathbf{u}_0 = \mathbf{q}$ and $\mathbf{u}_N = \mathbf{0}$. We rewrite the objective of $\mathcal{J}$ element-wise, using $\mathbf{f}_i$ and $\mathbf{u}_i$, as $\min_{\mathbf{q},\mathbf{F}} \mathbb{E}[\mathbf{x}^T\mathbf{x}] = \min_{\mathbf{q}} \sum_{i=1}^N \min_{\mathbf{f}_i} \sigma^2 q_i^2 + \sigma_s^2(q_i + \mathbf{f}_i^T\mathbf{u}_i)^2 + \sigma_s^2\|\mathbf{f}_i\|^2 \triangleq \min_{\mathbf{q}} \sum_{i=1}^N \min_{\mathbf{f}_i} J(q_i, \mathbf{f}_i)$. Solving for $\frac{\partial J(q_i,\mathbf{f}_i)}{\partial \mathbf{f}_i} = 0$ yields

$$\mathbf{f}_i = -\frac{q_i}{1 + \|\mathbf{u}_i\|^2}\mathbf{u}_i. \tag{10}$$

Plugging (10) in $J(q_i, \mathbf{f}_i) = \sigma^2 q_i^2 + \sigma_s^2 (q_i + \mathbf{f}_i^T \mathbf{u}_i)^2 + \sigma_s^2 \|\mathbf{f}_i\|^2$ leads to $\min_{\mathbf{f}_i} J(q_i, \mathbf{f}_i) = q_i^2 \left(\sigma^2 + \frac{\sigma_s^2}{1+\|\mathbf{u}_i\|^2}\right) \triangleq \tilde{J}(q_i, \mathbf{u}_i)$. Then, the overall minimization in (9) reduces to,

$$\min_{\mathbf{q}} \sum_{i=1}^{N} \tilde{J}_i(q_i, \mathbf{u}_i) = \sigma^2 \eta - \sigma_s^2 N + \sigma_s^2 \min_{\mathbf{q}} \sum_{i=1}^{N} \left(1 + \frac{q_i^2}{1+\|\mathbf{u}_i\|^2}\right).$$

We note that $1 + \frac{q_i^2}{1+\|\mathbf{u}_i\|^2} = \frac{1+\|\mathbf{u}_{i-1}\|^2}{1+\|\mathbf{u}_i\|^2}$, and by following the arithmetic mean–geometric mean inequality, we get $\sum_{i=1}^{N} \frac{1+\|u_{i-1}\|^2}{1+\|u_i\|^2} \geq N\left(\prod_{i=1}^{N} \frac{1+\|u_{i-1}\|^2}{1+\|u_i\|^2}\right)^{1/N}$. This lower bound is achieved by choosing $\mathbf{q}$ in order to satisfy $\frac{1+\|\mathbf{u}_{i-1}\|^2}{1+\|\mathbf{u}_i\|^2} = \delta$, yielding

$$\delta = (1+\eta)^{1/N}. \tag{11}$$

Thus, the minimum transmit power required to achieve the target SINR η in (9) is expressed as $\mathbb{E}[\mathbf{x}^T\mathbf{x}] = \sigma^2\eta - \sigma_s^2 N + \sigma_s^2 N(1+\eta)^{1/N}$. Since $\mathbb{E}[\mathbf{x}^T\mathbf{x}]$ is a monotonically increasing function of η, we can always find η that achieves

$$\mathbb{E}[\mathbf{x}^T\mathbf{x}] = \sigma^2\eta - \sigma_s^2 N + \sigma_s^2 N(1+\eta)^{1/N} = NP. \tag{12}$$

Since $q_i^2 = (1+\|\mathbf{u}_{i-1}\|^2) - (1+\|\mathbf{u}_i\|^2) = \delta^{N-i+1} - \delta^{N-i}$, the entries of $\mathbf{q}$ are given by

$$q_i = \sqrt{(\delta-1)\delta^{N-i}}, \quad i = 1, \ldots, N. \tag{13}$$

By plugging (13) in (10) and after algebraic manipulations, each entry of the optimal $\mathbf{F}$ can be written as

$$f_{i+j,i} = -(\delta-1)\delta^{-j/2}, \quad j = 1, \ldots, N-i. \tag{14}$$

Regarding $\lambda_{\min}(\mathbf{C})$ in (8), we have the following lemma.

Lemma 1. *The minimum eigenvalue of the interference-plus-noise covariance matrix* $\mathbf{C}$ *is* $\lambda_{\min}(\mathbf{C}) = \sigma_s^2 \delta^{-(N-1)} + \sigma^2$.

Proof. The proof is provided in Appendix A of [9]. ■

By Lemma 1, the θ encoding vector $\mathbf{g}$ is given by

$$g_i = \sqrt{\lambda_{\min}(\mathbf{C})} q_i = \sqrt{(\sigma_s^2 + \sigma^2 \delta^{N-1})(\delta-1)\delta^{-(i-1)}}. \tag{15}$$

C. SINR Bounds and Estimation Error Analysis

The minimum transmit power equality in (12) leads to

$$\frac{\eta}{N} = \frac{P}{\sigma^2} - \frac{\sigma_s^2}{\sigma^2}\left((1+\eta)^{1/N} - 1\right) \leq \frac{P}{\sigma^2}. \tag{16}$$

On the other hand, from (12)

$$NP \leq \sigma^2\eta - \sigma_s^2 N + \sigma_s^2 N\left(1 + \frac{\eta}{N}\right) = (\sigma^2 + \sigma_s^2)\eta, \tag{17}$$

where the inequality follows from the fact that $(1+\eta)^{1/N} \leq (1+\eta/N)$ for $N \geq 1$. Combining (16) and (17) yields

$$N\left(\frac{P}{\sigma^2 + \sigma_s^2}\right) \leq \eta \leq N\frac{P}{\sigma^2}. \tag{18}$$

The lower bound is achieved when $N = 1$. The upper bound becomes tighter as N grows, implying the proposed scheme effectively cancels out the interference as $N \to \infty$.

The likelihood $p(\hat{\theta}|\theta)$ is conditionally Gaussian distributed with mean $\mathbb{E}[\hat{\theta}|\theta] = \theta$ and variance $\mathbb{E}\left[(\hat{\theta}-\theta)^2 \mid \theta\right] = \frac{1}{\mathbf{g^T C^{-1} g}} = \frac{1}{\|\mathbf{q}\|^2} = \frac{1}{\eta}$, yielding

$$p\left(\hat{\theta} \mid \theta\right) = \sqrt{\frac{\eta}{2\pi}} \exp\left(-\frac{\eta(\hat{\theta}-\theta)^2}{2}\right). \tag{19}$$

An error occurs if there exists a symbol $\theta' \neq \theta$, $\theta', \theta \in \mathcal{C}$ such that $|\theta' - \hat{\theta}| \leq |\theta - \hat{\theta}|$. For M-ary pulse amplitude modulation (PAM), the distance between adjacent symbols is given by $d = 2\sqrt{\frac{3}{M^2-1}}$. Following the standard derivation for the average probability of error [10], P_e can be expressed as

$$P_e = \frac{M-1}{M}\int_{|\theta-\hat{\theta}| \geq \frac{d}{2}} p(\hat{\theta}|\theta)\, d\hat{\theta} = \frac{M-1}{M} \operatorname{erfc}\left(\sqrt{\frac{3\eta}{2(M^2-1)}}\right), \tag{20}$$

where $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}}\int_x^\infty e^{-t^2}\, dt$. For a fixed M, the proposed scheme ensures that the error probability in (20) decays exponentially with the blocklength N.

IV. Asymptotic Analysis

Leveraging the SINR bounds in (18) and the P_e in (20), we characterize the achievable rate of the proposed coding scheme. We first consider the case of noiseless interference channel, i.e., $z_i = 0$ in (1) and $\sigma^2 = 0$.

1) Noiseless Interference Channel: Under this channel, the capacity is expressed as $C_{\text{no-noise}} = \frac{1}{2}\log\left(1 + \frac{P}{\sigma_s^2}\right)$. From (12), the optimal SINR η is obtained as $\eta = \left(1 + \frac{P}{\sigma_s^2}\right)^N - 1$, implying the exponential growth of the SINR with the blocklength N. The probability of error is then expressed as $P_e = \left(1 - \frac{1}{2^{NR}}\right) \operatorname{erfc}\left(\sqrt{\frac{3}{2}\frac{2^{2NC_{\text{no-noise}}}-1}{2^{2NR}-1}}\right)$. Consequently, in the limit as $N \to \infty, P_e \to 0$ for $R < C_{\text{no-noise}}$ and $P_e \to 1$ for $R > C_{\text{no-noise}}$. Regarding the achievability of $R = C_{\text{no-noise}}$, we have the following lemma.

Lemma 2. *The proposed linear scheme achieves the channel capacity* $C_{\text{no-noise}}$ *when the strictly-casually known interference channel is noiseless.*

Proof. We choose M to scale as $M = \sqrt{N^{(1-\epsilon)}(1+\eta)}, \forall \epsilon > 1$. The achievable rate R is then computed as

$$\begin{aligned} R &= \lim_{N\to\infty} \frac{\log(M)}{N} \\ &= \lim_{N\to\infty} \frac{1}{2}\frac{(1-\epsilon)\log(N) + \sum_{i=1}^{N}\log(\delta)}{N} \\ &= \frac{1}{2}\log\left(1 + \frac{P}{\sigma_s^2}\right), \end{aligned} \tag{21}$$

where the second equality is due to (11). By (20), as $N \to \infty$, $P_e = \lim_{N\to\infty}\frac{M-1}{M}\operatorname{erfc}\left(\sqrt{\frac{3}{2}\frac{1}{N^{(1-\epsilon)}(1+\frac{1}{\eta}) - \frac{1}{\eta}}}\right) = 0$. This concludes the proof. ■

Lemma 2 states that as $N \to \infty$, the probability of error P_e vanishes double-exponentially, while the codebook cardinality $|\mathcal{C}| = M$ grows exponentially. Consequently, the achievable rate R converges to $C_{\text{no-noise}}$.

2) Noisy Interference Channel: When the channel is subject to both noise and interference as in (1), the follwoing holds.

Lemma 3. *In the presence of both noise and interference in* (1)*, the achievable rate R of the proposed linear coding scheme vanishes as $N \to \infty$.*

Proof. The achievable rate R is upper bounded by

$$\begin{aligned} NR &= H(\theta) \\ &\leq I\big(\theta;\hat{\theta}\big) + 1 + NRP_e && (22)\\ &= h\big(\hat{\theta}\big) - h\big(\hat{\theta}|\theta\big) + \epsilon_n && (23)\\ &\leq \frac{1}{2}\log\left(2\pi e\left(1+1/\eta\right)\right) - \frac{1}{2}\log\left(2\pi e\left(1/\eta\right)\right) + \epsilon_n && (24)\\ &\leq \frac{1}{2}\log\left(1 + N\frac{P}{\sigma^2}\right) + \epsilon_n, && (25)\end{aligned}$$

where (22) follows from the Fano's inequality applied to the message θ with blocklength N and message set size $M = 2^{NR}$. P_e is the average error probability in (20) with $P_e \to 0$ as $N \to \infty$. In (23), $\epsilon_N \triangleq 1+NRP_e$, which satisfies $\epsilon_N/N \to 0$ as $N \to \infty$. The inequality in (24) is due to (19) and the law of total variance, $\mathrm{Var}\big(\hat{\theta}\big) = \mathrm{Var}\big(\mathbb{E}[\hat{\theta}|\theta]\big) + \mathbb{E}\big[\mathrm{Var}\big(\hat{\theta}|\theta\big)\big] = 1 + \frac{1}{\eta}$. The inequality in (25) follows from (18). Consequently, from (25), $\lim_{N\to\infty} R \leq \frac{1}{2}\lim_{N\to\infty}\frac{\log\left(1+\frac{NP}{\sigma^2}\right)}{N} + \frac{\epsilon_N}{N} = 0$. Since R is non-negative, it concludes $R = 0$. ■

The results in Lemmas 2 and 3 share fundamental characteristics with the achievable rates established for linear feedback codes [4]–[7]. In the noiseless scenario, the availability of precise side information at the encoder enables the decoder to progressively refine prediction errors and achieve capacity. In the presence of noise, however, noise corrupts the encoded side information and prevents complete refinement, causing the asymptotic rate to vanish. While the zero achievable rate appears pessimistic, such asymptotic characterizations fail to capture the delicate behavior of rate and error performance in the finite-blocklength regime. This discrepancy motivates a rigorous characterization of the proposed coding scheme within the finite-blocklength regimes.

V. Finite-Blocklength Analysis

The asymptotic vanishing of the rate implies that for any $R > 0$, the error probability P_e remains bounded away from zero. This necessitates an analysis of the finite-blocklength performance to characterize the range of target error probabilities achievable by the proposed coding scheme for a fixed blocklength N.

Lemma 4. *For the proposed linear scheme with a fixed target error probability $\epsilon \in \left[\frac{1}{2}\mathrm{erfc}\Big(\sqrt{\frac{1}{2}\eta}\Big), 1\right)$, there exists some $\nu \in (0, N/3]$ such that the rate $R = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$ is achievable.*

Proof. From (18), for any $\rho \in \left[\frac{P}{\sigma^2+\sigma_s^2}, \frac{P}{\sigma^2}\right]$, we get $\eta = N\rho$. For an achievable R, there exists $\nu > 0$ such that $R = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$, in which the error probability is $P_e = \frac{M-1}{M}\mathrm{erfc}\Big(\sqrt{\frac{3}{2}\frac{N\rho}{2^{2NR}-1}}\Big) = \frac{M-1}{M}\mathrm{erfc}\Big(\sqrt{\frac{3}{2}\nu\rho}\Big) < \mathrm{erfc}\Big(\sqrt{\frac{3}{2}\nu\rho}\Big)$. Since $M \geq 2$, ν should satisfy $\nu \leq \frac{N}{3}$, implying $\epsilon \geq \frac{1}{2}\mathrm{erfc}\Big(\sqrt{\frac{\eta}{2}}\Big)$. Since $0 \leq \nu \ \leq \frac{N}{3}$, the proposed linear scheme achieves target error probability $\epsilon \in \Big[\frac{1}{2}\mathrm{erfc}\Big(\sqrt{\frac{\eta}{2}}\Big), 1\Big)$ with rate $R = \frac{\log(\lfloor\sqrt{1+\frac{N}{\nu}}\rfloor)}{N}$. This completes the proof. ■

Remark. *Lemma 4 establishes the characterizability of the finite-block-length regime for the proposed linear coding scheme. It shows that the proposed coding scheme can support a non-zero rate while maintaining a target error probability $\epsilon \in \big[\frac{1}{2}\mathrm{erfc}\big(\sqrt{\frac{1}{2}\eta}\big), 1\big)$. This motivates the selection of appropriate values for the transmission rate R, message size M, error probability ϵ, and blocklength N to ensure the effective operation of the scheme. Moreover, Lemma 4 also highlights that the interference power σ_s^2 and the noise power σ^2 are the primary factors affecting reliability performance.*

According to Shannon [3], access to strictly causal side information at the encoder does not increase channel capacity. In finite-blocklength regimes, we therefore define a baseline rate $R_{\text{no-SI},\epsilon}$, as the maximum achievable rate for a target error probability ϵ over an additive white Gaussian noise (AWGN) interference channel as in (1) without side information at the encoder, where the channel SNR is given by $\rho = \frac{P}{\sigma_s^2+\sigma^2}$. For a given target error probability ϵ, if the proposed scheme's achievable rate satisfies $R < R_{\text{no-SI},\epsilon}$, it is considered sub-optimal, as there exists an alternative coding scheme that supports a higher rate than R without utilizing interference side information. Conversely if $R \geq R_{\text{no-SI},\epsilon}$, the proposed scheme effectively leverages the available side information to outperform the no-side-information baseline.

A. Finite-blocklength Analysis for Binary Modulation

First, we characterize the achievable performance for any arbitrary finite blocklength N. For modulation order $M = 2$, the minimum achievable error probability can be determined, enabling us to evaluate whether the proposed scheme provides a reliability gain over the baseline.

Theorem 1. *For a modulation order of $M = 2$, we define the error probabilities as $\epsilon_0 = \frac{1}{2}\ \mathrm{erfc}\big(\sqrt{\frac{\eta}{2}}\big)$, and $\epsilon_{\text{no-SI}} = \frac{1}{2}\ \mathrm{erfc}\Big(\sqrt{\frac{NP}{2(\sigma_s^2+\sigma^2)}}\Big)$. Then, for every target error probability $\epsilon \in [\epsilon_0, \epsilon_{\text{no-SI}})$ and any blocklength $N > 1$, the proposed scheme achieves rate $\frac{1}{N}$, whereas any scheme without side information is limited to $R_{\text{no-SI},\epsilon} = 0$.*

Proof. First, we examine the AWGN interference channel without encoder side information. For $M = 2$, the message set contains two equiprobable symbols $\theta_1, \theta_2 \in \mathcal{C}$, encoded as equiprobable code vectors $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^N$. Bounded by the power constraint $\mathbf{x}_1^T\mathbf{x}_1 + \mathbf{x}_2^T\mathbf{x}_2 \leq 2NP$, the achievable error probability [10] is

$$P_{e,\text{no-SI}} = \frac{1}{2}\mathrm{erfc}\left(\sqrt{\frac{d_{\text{no-SI}}^2}{8(\sigma_s^2+\sigma^2)}}\right), \qquad (26)$$

where $d_{\text{no-SI}} = \|\mathbf{x}_1 - \mathbf{x}_2\|$ is the Euclidean distacne between the two codewords. Its squared is bounded by $d_{\text{no-SI}}^2 = \|\mathbf{x}_1 -$

$\mathbf{x}_2\|^2 = \|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2 - 2\mathbf{x}_1^T\mathbf{x}_2 \leq 2\|\mathbf{x}_1\|^2 + 2\|\mathbf{x}_2\|^2 \leq 4NP$. The maximum distance $d_{\text{no-SI}}$ is obtained when $\frac{\mathbf{x}_1^T\mathbf{x}_2}{\|\mathbf{x}_1\|\|\mathbf{x}_2\|} = -1$, meaning the signals are antipodal. The minimum achievable $P_{e,\text{no-SI}}$ is obtained as

$$P^{\star}_{e,\text{no-SI}} = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{NP}{2(\sigma_s^2+\sigma^2)}}\right). \tag{27}$$

This indicates that any target error probability $\epsilon < P^{\star}_{e,\text{no-SI}}$ is unattainable when $M = 2$. In other words, the achievable rate $R_{\text{no-SI},\epsilon} = 0$ for any $\epsilon < \epsilon_{\text{no-SI}}$, where $\epsilon_{\text{no-SI}} = P^{\star}_{e,\text{no-SI}}$.

For the proposed scheme, applying $M = 2$ to (20) yields the achieved probability of error,

$$P_e = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{1}{2}\eta}\right). \tag{28}$$

Leveraging the SINR bound in (18), P_e is bounded by $\frac{1}{2}\text{erfc}\left(\sqrt{\frac{NP}{2\sigma^2}}\right) \leq P_e \leq \frac{1}{2}\text{erfc}\left(\sqrt{\frac{NP}{2(\sigma^2+\sigma_s^2)}}\right)$. By setting $P_e = \epsilon$, it follows that for every $\epsilon \in [\epsilon_0, \epsilon_{\text{no-SI}})$, the proposed scheme achieves a strictly positive rate of $R = \frac{1}{N}$, whereas $R_{\text{no-SI},\epsilon} = 0$. This completes the proof. ■

Theorem 1 demonstrates the performance gain of our coding scheme for binary modulation $M = 2$. The proposed scheme successfully supports a positive transmission rate while achieving a lower target error probability, a performance level that is fundamentally unattainable by any scheme lacking access to interference side information.

B. Finite-Blocklength Analysis for Large-Blocklength Regime

When the blocklength N is sufficiently large, the framework of Polyanskiy *et al.* [8] can be used to further identify the specific blocklength regimes where the proposed coding scheme provides performance gains.

Definition 1. *[8, Definition 1] The channel dispersion V, which measures the stochastic variability relative to the channel capacity C, is defined as*

$$V = \limsup_{N\to\infty,\,\epsilon\to 0} \frac{1}{N}\left(\frac{NC - \log(M^{\star}(\epsilon, N))}{\sqrt{2}\text{erfc}^{-1}(2\epsilon)}\right)^2, \tag{29}$$

where $M^{\star}(\epsilon, N)$ is the maximum M to achieve the probability of error smaller than ϵ over a blocklength N.

Theorem 2. *(Normal Approximation) [8, Theorem 54] The maximum achievable rate R_ϵ for a target error probability ϵ over a finite blocklength N is given by*

$$R_\epsilon = \frac{\log\big(M^{\star}(\epsilon, N)\big)}{N} = C - \sqrt{\frac{2V}{N}}\,\text{erfc}^{-1}(2\epsilon) + O\Big(\tfrac{\log(N)}{N}\Big), \tag{30}$$

where C is the channel capacity and V is the dispersion defined in Definition 1. For an AWGN channel with signal-to-noise-ratio (SNR) ρ, the capacity is expressed as $C = \frac{1}{2}\log(1+\rho)$ and dispersion is $V = \frac{\rho}{2}\frac{\rho+2}{(\rho+1)^2}(\log e)^2$.

Polyanskiy's normal approximation [8] provides an estimate of $R_{\text{no-SI},\epsilon}$ as

$$\hat{R}_{\text{no-SI},\epsilon} = C - \sqrt{\frac{2V}{N}}\,\text{erfc}^{-1}(2\epsilon)\,, \tag{31}$$

where C and V share the same definitions as in Theorem 2, with the channel SNR $\rho = \frac{P}{\sigma^2+\sigma_s^2}$. However, $\hat{R}_{\text{no-SI},\epsilon}$ is an asymptotic expansion, and it generally differs from the exact finite-blocklength rate $R_{\text{no-SI},\epsilon}$. We characterize the gap by defining the approximation error $\Delta = R_{\text{no-SI},\epsilon} - \hat{R}_{\text{no-SI},\epsilon}$, which captures the higher-order terms neglected in the normal approximation and is dominated by terms of order $O\Big(\frac{\log(N)}{N}\Big)$. Accordingly, for a fixed, sufficiently large N and a target error probability ϵ, there exists a constant $\kappa > 0$ such that the exact finite-blocklength rate is bounded as

$$\hat{R}_{\text{no-SI},\epsilon} - \frac{\kappa\log(N)}{N} \leq R_{\text{no-SI},\epsilon} \leq \hat{R}_{\text{no-SI},\epsilon} + \frac{\kappa\log(N)}{N}. \tag{32}$$

This bound accounts for the residual gap between the normal approximation and the exact finite-blocklength rate, providing a reliable benchmark for evaluating the proposed scheme.

Theorem 3. *Consider $k = \frac{\sigma_s^2}{\sigma^2}$. Fix $\rho = \frac{P}{\sigma^2+\sigma_s^2}$ and κ as a constant. There exists N_0 such that for any $N > N_0$, there exists $k_0 > 1$ such that for $k > k_0$, we have $R > R_{\text{no-SI},\epsilon}$ to achieve a certain fixed target error probability ϵ.*

Proof. The proof is provided in Appendix B of [9]. ■

In summary, Theorem 3 shows that our coding scheme provides performance gains in high-interference regimes (i.e., $\sigma_s^2 > \sigma^2$). It demonstrates that the proposed scheme can support a higher transmission rate to achieve a sufficiently small target error probability compared to the baseline scenario, where the encoder has no access to interference side information.

VI. Conclusion

This paper proposed a linear coding scheme for Gaussian channels with strictly causal interference side information at the encoder. While we demonstrated that this scheme successfully achieves channel capacity in the noiseless scenario, the linear coding assumption inherently drives the asymptotic achievable rate zero in the presence of both interference and noise. Despite the asymptotic limitation, our finite-blocklength analysis reveals that the proposed scheme still offers distinct performance advantages. To overcome the vanishing-rate limitation, future work will focus on extending this framework by incorporating nonlinear or hybrid coding techniques. Motivated by the success of the deep learning-based codes developed by Kim *et al.* [11] for noisy feedback channels, exploring recurrent neural network (RNN) autoencoder architecture presents a highly promising avenue for channels with strictly causal side information at the encoder.

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Appendix A
Proof of Lemma 1

The eigenvalue decomposition (EVD) of $\mathbf{C} \in \mathbb{R}^{N\times N}$ is given by

$$\mathbf{C} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T, \tag{33}$$

where $\mathbf{Q} \in \mathbb{R}^{N\times N}$ is a unitary matrix whose columns are the eigenvectors of $\mathbf{C}$, i.e., $\mathbf{Q}^T\mathbf{Q} = \mathbf{Q}\mathbf{Q}^T = \mathbf{I}$, and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ contains the corresponding eigenvalues. Based on the EVD of $\mathbf{C}$, $\|\mathbf{q}\|^2$ is bounded by

$$\|\mathbf{q}\|^2 = \mathbf{g}^T\mathbf{C}^{-1}\mathbf{g} = \sum_{i=1}^{N}\frac{(\mathbf{Q}_i^T\mathbf{g})^2}{\lambda_i} \le \frac{\mathbf{g}^T\mathbf{g}}{\lambda_{\min}(\mathbf{C})}, \tag{34}$$

and maximizing $\|\mathbf{q}\|^2$ leads to (8). Thus, $\mathbf{q}$ is the eigenvector associated with the minimum eigenvalue of $\mathbf{C}$, i.e.,

$$\mathbf{C}\mathbf{q} = \sigma_s^2(\mathbf{F}+\mathbf{I})^T(\mathbf{F}+\mathbf{I})\mathbf{q} + \sigma^2\mathbf{q} = \lambda_{\min}(\mathbf{C})\mathbf{q}. \tag{35}$$

The i-th element of $(\mathbf{I}+\mathbf{F})(\mathbf{I}+\mathbf{F})^T\mathbf{q}$ can be expressed as

$$\begin{aligned}
&[(\mathbf{I}+\mathbf{F})(\mathbf{I}+\mathbf{F})^T\mathbf{q}]_i \\
&= q_i + \sum_{k=i+1}^{N} f_{k,i}q_k + \sum_{k=1}^{i-1} f_{i,k}\Big(q_k + \sum_{\ell=k+1}^{N} f_{\ell,k}q_\ell\Big) \\
&= q_i + \sum_{k=i+1}^{N} -(\delta-1)\delta^{-(k-i)}q_i \\
&\quad + \sum_{k=1}^{i-1} -(\delta-1)\Big(q_i + \sum_{\ell=k+1}^{N} -(\delta-1)\delta^{-(\ell-k)}q_i\Big)
\end{aligned} \tag{36}$$

$$\begin{aligned}
&= q_i + q_i\sum_{k=i+1}^{N} -(\delta-1)\delta^{-(k-i)} \\
&\quad + q_i\sum_{k=1}^{i-1} -(\delta-1)\Big(1 + \sum_{\ell=k+1}^{N} -(\delta-1)\delta^{-(\ell-k)}\Big) \\
&= q_i\Bigg[1-(\delta-1)\sum_{j=1}^{N-i}\delta^{-j} \\
&\quad - (\delta-1)\sum_{k=1}^{i-1}\Big(1-(\delta-1)\sum_{j=1}^{N-k}\delta^{-j}\Big)\Bigg] \qquad (37) \\
&= q_i\Bigg[1-(\delta-1)\sum_{j=1}^{N-i}\delta^{-j} - (\delta-1)\sum_{k=1}^{i-1}\delta^{-(N-k)}\Bigg] \qquad (38) \\
&= q_i\Bigg[1-(\delta-1)\Bigg(\sum_{j=1}^{N-i}\delta^{-j} + \sum_{j=N-i+1}^{N-1}\delta^{-j}\Bigg)\Bigg] \qquad (39) \\
&= q_i\Bigg[1-(\delta-1)\sum_{j=1}^{N-1}\delta^{-j}\Bigg] \\
&= q_i\,\delta^{-(N-1)}, \qquad (40)
\end{aligned}$$

where (36) is due to the equalities $q_k = \delta^{-(k-i)/2}q_i$ and $f_{k,i}q_k = -(\delta-1)\delta^{-(k-i)}q_i$, (37) is due to the equalities $\sum_{k=i+1}^{N}\delta^{-(k-i)} = \sum_{j=1}^{N-i}\delta^{-j}$ and $\sum_{\ell=k+1}^{N}\delta^{-(\ell-k)} = \sum_{j=1}^{N-k}\delta^{-j}$, (38) is due to the geometric sum equality $1-(\delta-1)\sum_{j=1}^{N-k}\delta^{-j} = 1-(1-\delta^{-(N-k)}) = \delta^{-(N-k)}$, and (39) follows from the fact that $\sum_{k=1}^{i-1}\delta^{-(N-k)} = \sum_{j=N-i+1}^{N-1}\delta^{-j}$ with $j = N-k$. Therefore, from (40), the minimum eigenvalue of $\mathbf{C}$ is given by $\lambda_{\min}(\mathbf{C}) = \sigma_s^2\delta^{-(N-1)} + \sigma^2$, and this completes the proof.

Appendix B
Proof of Theorem 3

We fix the target probability of error $\epsilon < 1/2$, and set $\rho = \frac{P}{\sigma^2+\sigma_s^2}$ as a constant. According to the SINR bound in (18), there exists N_0 such that for $N > N_0$,

$$\epsilon \ge \frac{1}{2}\mathrm{erfc}\left(\sqrt{\frac{N\rho}{2}}\right) \ge \frac{1}{2}\mathrm{erfc}\left(\sqrt{\frac{\eta}{2}}\right). \tag{41}$$

In the high-interference regime where $k > 1$, the SINR η is characterized by $\eta = \left(1+\frac{P}{\sigma_s^2}-\frac{\eta}{kN}\right)^N - 1$, which yields the following lower bound

$$\eta \ge \frac{k\left[\left(1+\frac{P}{\sigma_s^2}\right)^N - 1\right]}{k+\left(1+\frac{P}{\sigma_s^2}\right)^{N-1}}. \tag{42}$$

The detailed derivation of (42) is provided in Appendix C. By utilizing the error probability expression in (20), the achievable rate R for a given target error probability ϵ satisfies

$$\begin{aligned}
R &= \frac{1}{2N}\log\left(\frac{3\eta}{2(\mathrm{erfc}^{-1}(\frac{M}{M-1}\epsilon))^2}+1\right) \\
&> \frac{1}{2N}\log(\eta) - \frac{\log(\mathrm{erfc}^{-1}(\epsilon))}{N} \\
&\ge \frac{1}{2}\log\Big(1+\frac{P}{\sigma_s^2}\Big) - \frac{1}{2N}\log\left(\frac{1+\frac{(1+\frac{P}{\sigma_s^2})^{N-1}}{k}}{1-(1+\frac{P}{\sigma_s^2})^{-N}}\right) \\
&\quad - \frac{\log(\mathrm{erfc}^{-1}(\epsilon))}{N},
\end{aligned} \tag{43}$$

where the last step is due to (42) and $\log\left((1+\frac{P}{\sigma_s^2})^N - 1\right) = \log\left((1+\frac{P}{\sigma_s^2})^N\right) + \log\left(1-(1+\frac{P}{\sigma_s^2})^{-N}\right)$. For convenience, we denote $A(k) = \frac{1}{2}\log\left(\frac{1+\frac{\left(1+\rho\left(1+\frac{1}{k}\right)\right)^{N-1}}{k}}{1-\left(1+\rho\left(1+\frac{1}{k}\right)\right)^{-N}}\right)$, and $C = \frac{1}{2}\log(1+\rho)$. The limit of $A(k)$ as $k\to\infty$ is given by

$$G(N) = \lim_{k\to\infty}A(k) = \frac{1}{2}\log\left(\frac{1}{1-(1+\rho)^{-N}}\right). \tag{44}$$

Therefore, for any $\mathcal{E} > 0$, there exists a $k_0(N)$, such that for $k > k_0(N)$, we have

$$|A(k) - G(N)| \le \mathcal{E}.$$

We set $\mathcal{E} = \frac{G(N)}{2}$ and then $A(k)$ is upper bounded by $A(k) \leq \frac{3G(N)}{2}$. Since $G(N)$ is decreasing with N, there exists N_1 such that for any $N > N_1$, $A(k) < 2$ for $k > k_0(N)$. Thus from (43),

$$R > C - \frac{2}{N} - \frac{\log(\mathrm{erfc}^{-1}(\epsilon))}{N}. \tag{45}$$

From (32), since $\epsilon < \frac{1}{2}$, the baseline rate $R_{\text{no-SI},\epsilon}$ satisfies

$$R_{\text{no-SI},\epsilon} \leq C - \sqrt{\frac{2V}{N}}\mathrm{erfc}^{-1}(2\epsilon) + \kappa\frac{\log N}{N}. \tag{46}$$

According to (45) and (46), the gap between the two rates is lower bounded as $N(R - R_{\text{no-SI}}, \epsilon) \geq \sqrt{2VN}\,\mathrm{erfc}^{-1}(2\epsilon) - \kappa\log(N) - \log\big(\mathrm{erfc}^{-1}(\epsilon)\big) - 2$. We now define a continuous function $B(N)$ to represent this lower bound as

$$B(N) = \sqrt{2VN}\mathrm{erfc}^{-1}(2\epsilon) - \kappa\log(N) - \log\big(\mathrm{erfc}^{-1}(\epsilon)\big) - 2. \tag{47}$$

If $B(N) > 0$, it guarantees $R > R_{\text{no-SI},\epsilon}$ when N is sufficiently large. To analyze its behavior, we differentiate $B(N)$ with respect to N and set the derivative equal to zero: $B'(N) = \frac{\mathrm{erfc}^{-1}(2\epsilon)\sqrt{2V}}{2\sqrt{N}} - \frac{\kappa}{N} = 0$. The stationary point is obtained by $N^\star = \frac{2\kappa^2}{V\left(\mathrm{erfc}^{-1}(2\epsilon)\right)^2}$. This implies that for $N > N^\star$, $B(N)$ is monotonically increasing. Consequently, there exists an $N_2 > N^\star$ such that for $N > N_2$, $B(N) > 0$. In conclusion, for blocklength $N > \max(N_0, N_1, N_2)$, there exists $k_0(N)$ such that for $k > k_0(N)$, $R_{\text{no-SI},\epsilon} < R$. This completes the proof.

Appendix C
Derivations of SINR for $\sigma_s^2 > \sigma^2$

We defne $p = \frac{P}{\sigma_s^2}$. For $k = \frac{\sigma_s^2}{\sigma^2} > 1$, we have

$$\begin{aligned} \eta + 1 &= \left(1 + p - \frac{\eta}{kN}\right)^N \\ &= (1+p)^N\left(1 - \frac{\eta}{kN(1+p)}\right)^N \\ &\geq (1+p)^N\left(1 - \frac{\eta}{k(1+p)}\right), \end{aligned} \tag{48}$$

where the inequality in the last step is due to Bernoulli's inequality $(1 - \frac{\eta}{kN(1+p)})^N \geq 1 - \frac{\eta}{k(1+p)}$ for $\frac{\eta}{kN(1+p)} \in [0,1]$. The inequality in (48) can be rewritten as

$$\eta\left(1 + \frac{(1+p)^{N-1}}{k}\right) \geq (1+p)^N - 1. \tag{49}$$

Substituting $p = \frac{P}{\sigma_s^2}$ pack into the equation, we obtain the following bound for η,

$$\eta \;\geq\; \frac{\left(1 + \frac{P}{\sigma_s^2}\right)^N - 1}{1 + \frac{\left(1+\frac{P}{\sigma_s^2}\right)^{N-1}}{k}}. \tag{50}$$